Mlp
Benchmarks

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1. Training Performance

- **Training Time:** 18 minutes for 3 epochs.
- **Metrics:** Cross-entropy loss, Accuracy.
- **Dataset Size:** 700,000 FEN strings, individually labeled with their corresponding board states.
- **Convergence:** The cost function demonstrates consistent reduction throughout the training process.

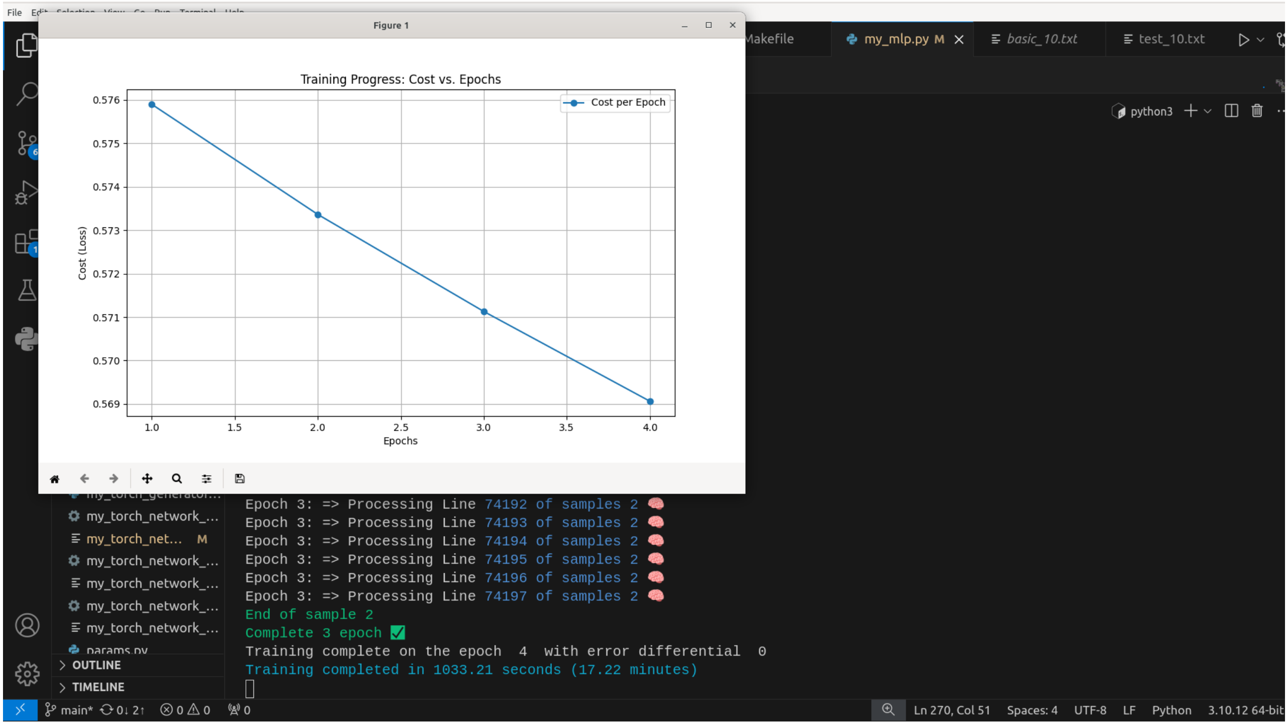

Figure 1 : Cost reduction during training

Figure 2 : Execution time

Accuracy: An initial success rate of 70% was achieved on basic evaluation.

Figure 3 : *Taux de réussite*

Prediction Example

- **Input (FEN):** rnb1kbnr/pppp1ppp/8/4p3/6Pq/5P2/PPPPP2P/RNBQKBNR w KQkq - 1 3
- **Output:** Check

3. Overfitting Mitigation

To ensure model generalization and prevent overfitting, the following strategies were implemented:

- **Data Consolidation:** All datasets were merged into a single comprehensive file.
- **Shuffling:** Data was partitioned into samples and randomized to enhance the AI's predictive accuracy.
- **Learning Rate Decay:** The learning rate was reduced by 5% per epoch to maintain training stability, prevent gradient explosion, and avoid convergence toward suboptimal local minima.

4. Configuration & Architecture

The neural network generation is governed by a configuration file with the following specifications:

- **Input Dimension:** Number of network inputs.
- **Layer Count:** Total number of layers.
- **Neuron Density:** Number of neurons per layer.
- **Activation Mapping:** Specific activation function assigned to each layer.
- **Hyperparameters:** Initial learning rate value.

The output is a .nn model file containing:

- **Architecture Metadata:** Inputs, layers, and neurons per layer.
- **Parameter Dictionary ("param"):** * **Weights (Wn):** Layer-specific weight matrices.
  - **Functions (Fn):** Activation functions associated with each layer.
  - **Biases (Bn):** Layer-specific bias vectors.

5. Design Justifications

- **Activation Functions:**
    - **ReLU:** Utilized for its efficiency in handling non-linearities and mitigating vanishing gradient issues.
    - **Softmax:** Implemented at the output layer to provide a probability distribution across multiple categorical states.
- **Optimization:**
    - **Scheduled Decay:** A 5% per-epoch learning rate reduction was applied to stabilize the training trajectory and ensure smoother convergence.